

UNITS AND MEASUREMENT

Class 11 Physics — Complete Comprehensive Concept Notes & Formula Revision

CBSE Class 11

JEE Main & Advanced

NEET Physics

1. INTRODUCTION & PHYSICAL QUANTITIES

Physics is a quantitative science based on measurement of physical quantities. A **Physical Quantity** is any property of a material or system that can be quantified and measured using a standard unit.

$$\text{Physical Quantity (Q)} = \text{Numerical Value (n)} \times \text{Unit (u)} \Rightarrow Q = n_1 u_1 = n_2 u_2 = \text{constant}$$

Classification of Physical Quantities

Category	Definition	Examples
Fundamental (Base) Quantities	Quantities that are independent of other quantities and cannot be defined in terms of other physical quantities.	Length, Mass, Time, Temperature, Electric Current, Luminous Intensity, Amount of Substance.
Derived Quantities	Quantities derived from combination and algebraic manipulation of fundamental quantities.	Velocity (m/s), Force ($kg \cdot m/s^2$), Work (J), Pressure (Pa).
Supplementary Quantities	Geometrical non-dimensional quantities added to complete the SI system.	Plane Angle (Radian), Solid Angle (Steradian).

2. THE SI SYSTEM OF UNITS

The *International System of Units (SI)* defines 7 base units and 2 supplementary units:

Base Quantity	SI Base Unit	Symbol	Dimension
Length	Metre	m	[L]
Mass	Kilogram	kg	[M]
Time	Second	s	[T]
Electric Current	Ampere	A	[I] or [A]
Thermodynamic Temperature	Kelvin	K	[K] or [θ]
Amount of Substance	Mole	mol	[mol]
Luminous Intensity	Candela	cd	[cd]

Supplementary Units

- **Plane Angle ($d\theta$):** Ratio of arc length (ds) to radius (r). Unit: **radian (rad)**.

$$d\theta = ds / r \text{ (Dimensionless: } [M^0 L^0 T^0])$$

- **Solid Angle ($d\Omega$):** Ratio of intercepted area (dA) to square of radius (r^2). Unit: **steradian (sr)**.

$$d\Omega = dA / r^2 \text{ (Dimensionless: } [M^0 L^0 T^0])$$

3. DIMENSIONAL ANALYSIS

The **Dimensions** of a physical quantity are the powers (or exponents) to which the fundamental base quantities are raised to represent that quantity.

Principle of Homogeneity of Dimensions

According to this principle, a physical equation is dimensionally correct if and only if the dimensions of each term on both sides of the equation are identical.

If $A + B = C$, then $[A] = [B] = [C]$.

Note: Trigonometric, logarithmic, and exponential functions, as well as pure numbers, are strictly dimensionless.

Dimensions of Important Physical Quantities

Physical Quantity	Formula / Relation	Dimensional Formula	SI Unit
Velocity / Speed	$v = \text{distance} / \text{time}$	$[M^0 L T^{-1}]$	m/s
Acceleration	$a = \Delta v / t$	$[M^0 L T^{-2}]$	m/s ²
Force / Weight	$F = m \times a$	$[M L T^{-2}]$	N (Newton)
Work / Energy / Torque	$W = F \times d$	$[M L^2 T^{-2}]$	J (Joule)
Power	$P = \text{Work} / \text{time}$	$[M L^2 T^{-3}]$	W (Watt)
Pressure / Stress	$P = \text{Force} / \text{Area}$	$[M L^{-1} T^{-2}]$	Pa (Pascal)
Universal Gravitational Constant (G)	$F = G m_1 m_2 / r^2$	$[M^{-1} L^3 T^{-2}]$	N·m ² /kg ²
Planck's Constant (h)	$E = h \nu$	$[M L^2 T^{-1}]$	J·s
Surface Tension	$T = \text{Force} / \text{Length}$	$[M L^0 T^{-2}]$	N/m
Coefficient of Viscosity (η)	$F = \eta A (dv/dx)$	$[M L^{-1} T^{-1}]$	Pa·s / Poiseuille

Applications of Dimensional Analysis

- Checking Dimensional Consistency:** Verifying if a given physical relationship is dimensionally correct.
- Conversion of Units across Systems:** Using $n_2 = n_1 [M_1/M_2]^a [L_1/L_2]^b [T_1/T_2]^c$ to convert values between SI, CGS, MKS.
- Deducing Relations among Physical Quantities:** Deriving functional formulas when dependencies are known (e.g., Simple Pendulum time period $T = 2\pi \sqrt{l/g}$).

Limitations of Dimensional Analysis

- Does not give information about dimensionless constants (e.g., 2π , numerical factors).
- Fails for equations involving trigonometric, exponential, or logarithmic functions.
- Cannot derive relations containing more than 3 unknown physical variables if relying on M, L, T alone.
- Cannot determine whether a physical quantity is a scalar or vector.

4. ERRORS IN MEASUREMENT

Every measurement involves some uncertainty, known as an **Error**.

$$\text{Error} = \text{True Value} - \text{Measured Value}$$

Types of Errors

- **Systematic Errors:** Tend to be in one direction (either positive or negative).
 - *Instrumental Errors:* Faulty design/zero error of instrument.
 - *Imperfection in Technique:* External parameters like thermal radiation, buoyancy.
 - *Personal Errors:* Parallax or improper setting of apparatus by observer.
- **Random Errors:** Occur irregularly in magnitude and sign due to unpredictable fluctuations. Minimised by taking arithmetic mean of multiple observations.

Mathematical Formulation of Errors

Let $a_1, a_2, \dots, a_n$ be n measured values of a quantity:

1. Mean (True) Value:

$$a_{\text{mean}} = (a_1 + a_2 + \dots + a_n) / n = (1/n) \sum a_i$$

2. Absolute Error: Magnitude of difference between true value and individual measured value:

$$\Delta a_i = |a_{\text{mean}} - a_i|$$

3. Mean Absolute Error:

$$\Delta a_{\text{mean}} = (1/n) \sum |\Delta a_i|$$

Final result is expressed as: $a = a_{\text{mean}} \pm \Delta a_{\text{mean}}$

4. Relative (Fractional) Error:

$$\text{Relative Error} = \Delta a_{\text{mean}} / a_{\text{mean}}$$

5. Percentage Error:

$$\text{Percentage Error} = (\Delta a_{\text{mean}} / a_{\text{mean}}) \times 100\%$$

Propagation / Combination of Errors

Operation	Physical Relation	Maximum Absolute / Relative Error
Sum / Difference	$Z = A \pm B$	$\Delta Z = \Delta A + \Delta B$ (Absolute errors add up)
Product / Division	$Z = A \times B$ or $Z = A / B$	$(\Delta Z / Z) = (\Delta A / A) + (\Delta B / B)$ (Relative errors add up)
Power of a Quantity	$Z = A^p B^q / C^r$	$(\Delta Z / Z) = p(\Delta A / A) + q(\Delta B / B) + r(\Delta C / C)$

5. SIGNIFICANT FIGURES & ROUNDING OFF

Significant figures in a measured quantity indicate the precision of measurement and include all reliably known digits plus the first uncertain digit.

Rules for Counting Significant Figures

1. All non-zero digits are significant (e.g., **1234** has 4 SF).
2. All zeroes between two non-zero digits are significant (e.g., **1005** has 4 SF).
3. Leading zeroes (before non-zero digits) are NEVER significant (e.g., **0.0025** has 2 SF).
4. Trailing zeroes in a number WITHOUT a decimal point are NOT significant (e.g., **1500** has 2 SF).
5. Trailing zeroes in a number WITH a decimal point ARE significant (e.g., **3.500** has 4 SF, **0.050** has 2 SF).
6. Exact numbers (like π , counting numbers) have an infinite number of significant figures.

Arithmetic Operations with Significant Figures

- **Addition & Subtraction:** The final result should retain as many decimal places as present in the number with the *least number of decimal places*.
- **Multiplication & Division:** The final result should retain as many significant figures as present in the quantity with the *least number of significant figures*.

6. MEASURING INSTRUMENTS (PRACTICAL PHYSICS)

A. Vernier Caliper

Used to measure small linear dimensions (diameter, depth, length).

Least Count (LC):

$$LC = 1 \text{ MSD} - 1 \text{ VSD}$$
$$LC = \text{Value of 1 MSD} / \text{Total Vernier Divisions}$$

Total Reading (TR):

$$TR = MSR + (VSR \times LC) - (\text{Zero Error})$$

- **Positive Zero Error:** Vernier zero is to the right of Main zero (Subtract error).
- **Negative Zero Error:** Vernier zero is to the left of Main zero (Add error).

B. Screw Gauge (Micrometer)

Used to measure thin wire thickness, glass sheet thickness, etc.

Pitch: Distance moved on main scale per full revolution of thimble.

Least Count (LC):

$$LC = \text{Pitch} / \text{Total Circular Scale Divisions}$$

Total Reading (TR):

$$TR = PSR + (CSR \times LC) - (\text{Zero Error})$$

- **Pitch:** Usually 1 mm or 0.5 mm.
- Standard LC is typically 0.01 mm or 0.001 cm.

7. QUICK SUMMARY FORMULA SHEET

Unit Transformation:

$$n_1 u_1 = n_2 u_2$$

Combination of Error (Power Rule):

$$\text{If } Z = A^a B^b / C^c, \text{ then } \Delta Z/Z = a(\Delta A/A) + b(\Delta B/B) + c(\Delta C/C)$$

Vernier Caliper LC:

$$LC = 1 \text{ MSD} - 1 \text{ VSD} = (1 - N_{\text{VSD}}/N_{\text{MSD}}) \times \text{MSD}$$

Screw Gauge LC:

$$LC = \text{Pitch} / \text{Total Circular Scale Divisions}$$

Parallax Method (Distance):

$$D = b / \theta \text{ (where } b = \text{basis, } \theta = \text{parallax angle in radians)}$$